

Mangrove Vulnerability Assessment of Pichavaram Using Remote Sensing and GIS-Based Multi-Criteria Analysis

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Abstract

Mangrove ecosystems serve as an important ecological buffer along the coastlines of India. It protects the coast and helps to capture the carbon as blue carbon. The Pichavaram mangrove forest in Tamil Nadu is among the largest contiguous mangrove stands in India. This study lays out a detailed geospatial multi-criteria vulnerability assessment of the Pichavaram mangrove ecosystem, using a mix of satellite remote sensing data, land use & landcover change detection, vegetation index analysis, patch-level fragmentation analysis, and vulnerability assessment. Four main thematic layers were created and combined through a Weighted Overlay Analysis (WOA) in a GIS setting by using Mangrove NDVI from Landsat 8 OLI imagery 2025, a mangrove transition layer formed from bi-temporal LULC classifications 2015 and 2025, a Fragmentation Vulnerability Map based on patch area analysis, Core Area Analysis for measuring interior habitat extent, and a Digital Elevation Model (DEM) for topographic exposure. Weights for each layer came from expert judgment and a review of existing literature. The resulting Mangrove Vulnerability Assessment separates the Pichavaram area into three vulnerability zones such as Highly Vulnerable, Vulnerable, and Low Vulnerable. The results show that about 8% of the assessed mangrove area falls into the High to Very High vulnerability classes, mostly around the forest's edges near agricultural fields, aquaculture ponds, and sedimentation-prone channels. In contrast, core interior patches displayed low fragmentation and higher NDVI values, which corresponded Low vulnerability. This study highlights the usefulness of merging multi-source geospatial data with multi-criteria decision analysis (MCDA) for precise mangrove vulnerability mapping, offering a data-driven framework to guide conservation planning, ecosystem restoration priorities, and policy actions for the Pichavaram mangrove ecosystem.

Keywords: *Mangrove vulnerability; Pichavaram; Fragmentation; GIS; Remote sensing; multi-criteria analysis.*

Introduction

Mangrove forests are among the most productive and ecologically diverse biomes on Earth, occupying the intertidal zones of tropical and subtropical coastlines across approximately 137,760 km² globally (Giri et al., 2011). These ecosystems deliver a disproportionately large suite of ecosystem services relative to their spatial extent, including blue carbon storage, shoreline stabilisation, reduction of wave energy and storm surge, water quality regulation, and support for fisheries productivity (Donato et al., 2011; Mcleod et al., 2011; Spalding et al., 2014; Murdiyarswo et al., 2015). Despite their ecological and socioeconomic value, mangrove forests have experienced rapid and widespread degradation at a global scale over the past half century. Between 1980 and 2005, an estimated 20-35% of the world's mangrove cover was lost, primarily driven by aquaculture expansion, coastal agriculture, urban development, unsustainable harvesting, and hydrological alterations (Valiela et al., 2001; FAO, 2007). They are groups of trees that can grow in estuaries and form a transition zone between land and sea (Duke et al., 2007). Global mangrove deforestation shows a tremendous decline of between 0.16% and 0.39% per year (Hamilton & Casey, 2016). India, home to nearly 4,975 km² of mangrove cover has witnessed significant decline and fragmentation of its mangrove ecosystems along the eastern and western coasts (Bhatt & Bhatt, 2020). mangrove ecosystems are important habitats, especially in developing countries and will continue to be exploited in 2025, well they have seen as a valuable resource to be managed on a sustainable basis. After 2025, the future of mangroves will depend on technological and ecological advances (Alongi, 2002). While the evolution of mangrove forest is identified through remote sensing techniques from 1956 itself (Wang et al., 2019).

The Mangroves are considered to behave like a natural barrier against ocean dynamics along the shoreline. Their ability to protect shoreline and inland areas from natural hazards (Kuenzer et al., 2011; Alongi, 2008). The Pichavaram mangrove forest, situated on the Coromandel Coast of Tamil Nadu, is one of the most ecologically significant and well-studied mangrove ecosystems in India. It is the second largest mangrove forest in India and is recognised as a Ramsar wetland. Traditional ground-based monitoring is constrained by the spatial extent and inaccessibility of many mangrove habitats. While Satellite remote sensing provides an effective and cost-efficient alternative, enabling synoptic observations of

vegetation condition, land-cover dynamics, and terrain properties at regular temporal intervals (Pham et al., 2019; Wang et al., 2019). GIS-based multi-criteria decision analysis (MCDA), particularly Weighted Overlay Analysis (WOA), offers a structured and spatially explicit framework for synthesising heterogeneous raster datasets into integrated vulnerability indices (Malczewski, 1999).

Several studies have applied geospatial MCDA techniques to assess the vulnerability of coastal ecosystems along with coastal vulnerability index (CVI) (Nandi et al., 2016; Pantusa et al., 2018; Pendleton et al., 2012). However, the integration of vegetation spectral health (NDVI), decadal LULC transition, landscape fragmentation vulnerability, core area analysis, and elevation-based inundation exposure into a single composite vulnerability layer for the Pichavaram mangrove system has not been comprehensively undertaken. This study addresses this gap by constructing a Mangrove Vulnerability Zones through the weighted overlay of four spatially derived thematic layers thereby providing a fine-scale vulnerability map (Figure 1). The objective of this study is to map and quantify decadal LULC change and mangrove transition dynamics between 2015 and 2025, to derive and classify mangrove NDVI as an indicator of current vegetation health, to analyse landscape fragmentation and core-patch geometry of the mangrove ecosystem, to integrate DEM-derived topographic exposure with biotic and anthropogenic stress layers and to produce a composite Mangrove Vulnerability Index map for the Pichavaram region.

Study area

The Pichavaram mangrove forest is located on the southeastern coast of India in the Cuddalore district of Tamil Nadu, between latitudes 11°30'8.10"N - 11°21'40.89"N and longitudes 79°46'50.37"E - 79°49'47.35" E. It lies at the confluence of the Vellar and Coleroon rivers, two significant distributaries of the Cauvery river system. The forest is positioned approximately 3 km from the town of Chidambaram and is bordered by the Bay of Bengal to the east, and the study area is defined by deriving 10 km buffer from the coastline of Pichavaram forest. The areas between the two Rivers have brackish water with mangrove vegetation (Sankari et al., 2015). The dominant mangrove species assemblages at Pichavaram include *Avicennia marina*, *Avicennia officinalis*, *Rhizophora apiculata*, *Rhizophora mucronata*, *Aegiceras corniculatum*, *Excoecaria agallocha*, and *Bruguiera cylindrica*, distributed in distinct zonation patterns influenced by salinity gradients, inundation frequency, and sediment texture (Selvam & Kathiresan, 2010). *Rhizophora* species tend to occupy the seaward fringe and creek margins,

while *Avicennia* species dominate the landward transition zones. The ecosystem supports rich avian biodiversity, including migratory and resident water birds, and constitutes important nursery grounds for commercially significant fin and shellfish species (Kathiresan & Bingham, 2001).

The forest was designated a Reserved Forest under the Tamil Nadu Forest Act and is managed by the Tamil Nadu Forest Department. It was declared a Ramsar Wetland of International Importance in 2022, reflecting its global biodiversity and hydrological significance. However, the surrounding landscape is densely populated and subject to intensive land use. Aquaculture ponds, particularly for shrimp farming, occupy significant areas adjacent to the mangrove fringe. Agricultural fields under paddy and cash crops border the landward edge (Prabhadevi & Reddy, 2012). These multiple anthropogenic stressors, superimposed on natural climate and hydrological variability, render Pichavaram a compelling and policy-relevant site for vulnerability assessment.

This study aims to assess the vulnerability of the Pichavaram mangrove forest for the year 2025 using Landsat 8 OLI imagery. Mangrove extents for 2015 and 2025 were extracted through Land Use/Land Cover classification and compared to identify areas of mangrove gain, stability, and loss. The mangrove transition map (2015-2025), along with additional parameters including the 2025 Normalized Difference Vegetation Index (NDVI), fragmentation vulnerability derived from patch and core area analyses, and Digital Elevation Model (DEM) data, were integrated using a GIS-based multi-criteria Analysis to delineate and assess mangrove vulnerability zones.

Methodology

Data sources and pre-processing

Multi-temporal satellite imagery was acquired such as Landsat 8 OLI (2015) and Landsat 8 OLI (2025), both obtained from the USGS Earth Explorer. Cloud-free scenes were selected for both time periods corresponding to the post-monsoon dry season (January-March) to minimise atmospheric effects (Roy et al., 2014). Landsat 8 OLI imagery Level 2 (surface reflectance) was utilised for 2025 NDVI computation at 30 m spatial resolution and used for the LULC 2025 baseline. Pre-processing steps included layer stacking of relevant spectral bands. Geometric correction was applied to align all datasets to a common spatial reference framework (WGS 1984 / UTM Zone 44N). A 30 m spatial resolution Digital Elevation Model was derived

from the SRTM DEM to account for systematic underestimation of low-relief terrain (Farr et al., 2007).

LULC Classification and Mangrove Transition (2015-2025)

Land use and land cover classification was conducted using a supervised classification with Maximum Likelihood Classification algorithm in QGIS (Richards, 2013; Congalton & Green, 2009). Training samples for each class were digitised from Landsat 8 OLI Level 2 imagery. Five LULC classes were identified such as Mangroves, Fallow Land, Crop Land, Sandy Area, Waterbody for 2015 (Figure 2, Figure 3) and 2025 respectively (Figure 3). Mangrove extraction was done for the year 2015 and 2025 from the LULC (Figure 4), which has determined the mangrove expansion in terms of area (ha) over 96.6 ha from 2015 to 2025. Where in 2015 it showed 1373.9 ha and for 2025 it consists of 1470.5 ha as total mangrove area. Mangrove transition analysis was conducted through post-classification change detection. A transition matrix was constructed quantifying the area of conversion between each pair of LULC classes across the two time periods. Pixels that transitioned from Dense Mangrove to non-mangrove classes (Crop Land, Fallow Land, Waterbody, Sandy Area) were flagged as loss areas and assigned high vulnerability scores. The pixels showing gain or stability in mangrove cover were assigned low transition vulnerability values. The resulting transition layer was reclassified into Non-Mangrove, Mangrove Stable, Mangrove Loss, Mangrove Gain based on the magnitude and direction of change (Figure 5).

Spatial Pattern of Mangrove Vegetation Health Using NDVI (2025)

The Normalised Difference Vegetation Index (NDVI) was computed from Landsat 8 OLI imagery 2025 using the standard formulation (Rouse et al., 1974; Tucker, 1979).

$$NDVI = (NIR - Red) / (NIR + Red)$$

Where NIR corresponds to Landsat 8 OLI Band 5 and Red to Band 4 both at 30 m resolution. The NDVI layer was clipped to the mangrove extent boundary (Figure 6) and reclassified into five classes based on natural breaks. Low NDVI values indicate canopy thinning, dieback, or degradation and were assigned higher vulnerability, while high NDVI values indicative of dense, vigorous canopy were assigned low vulnerability (Flores-de-Santiago et al., 2012; Heumann, 2011).

Patch Area and Core Area Analyses

Patch area analysis was done for the year 2015 and 2025 that determined the total no of patches for 2015 and 2025 as 1393 and 932 patch counts (Figure 7). Then the mean patch size was calculated by dividing total mangrove area with total no of patches for respective years. Which resulted mean patch size for 2015 and 2025 as 0.99 ha and 1.58 ha. This increase coupled with the reduction in patch count, suggests a trend toward larger and more contiguous mangrove patches, indicating a decrease in fragmentation over the study period. Where very small patches are decreased with increase in large patches that leads to the increase of core area coverage and percentage along with decrease of patch counts from 1393 to 932 that turns into decrease in fragmentation between 2015 and 2025. The observed increase in mean patch size and decrease in patch count indicate improved patch connectivity and reduced fragmentation, which generally enhances the ecological resilience of the mangrove ecosystem (Halpern et al., 2007; Nagelkerken et al., 2015). Meanwhile Core area was defined as the interior habitat zone of each mangrove patch attained with buffer -100 m from the edge (Laurance et al., 2002). Then the core area analysis was computed for each patch of 2015 and 2025 mangrove layer with the help of zonal statistics tool. Which resulted in the core area coverage of 237.12 ha for the year 2015 and 330.32 ha for 2025 in which those areas are falls on the patch class of very large, large, medium along with core area percentage 17.26% for 2015 and 22.46% for 2025 (Figure 8).

Fragmentation Vulnerability Analysis

Fragmentation of mangrove areas refers to the sub-division of a continuous landscape into smaller units which can be characterized based on patch size, edge, and shape modification (Kanniah et al., 2021). Forest fragmentation is the process of breaking up large, continuous areas of forest into smaller and isolated patches, usually due to human activities such as roads, agriculture, urbanization, or logging (Roy et al., 2025). Landscape fragmentation was assessed by using patch size fragmentation analysis through patch class which computed in QGIS on the 2025 mangrove LULC binary layer which was converted into a vector layer to do the patch class analysis. The patch classes included with patch counts is used to define the fragmentation score of Pichavaram mangrove system. Patches with small area (ha) coverage, irregular shapes, and high edge density were considered more vulnerable to edge effects. Based on the patch class of 2025, fragmentation score was determined then it reclassified into five vulnerability classes (Figure 9). Very small patches with high fragmentation score were assigned Very High

vulnerability, reflecting their susceptibility to total loss under continued disturbance. Large, contiguous patches with high score were assigned Very Low vulnerability.

DEM Based Topographic Vulnerability

The SRTM DEM was used to derive topographic vulnerability indicators relevant to tidal inundation risk and sea-level rise exposure (Dasgupta et al., 2009; Tran et al., 2008). The DEM was reclassified into five vulnerability classes, ranging from very low to very high vulnerability. Most of the study area falls within the high and very high vulnerability classes, indicating predominantly low-lying terrain. These areas are more prone to tidal flooding, storm surges, and sea-level rise impacts (Nicholls & Cazenave, 2010). In contrast, only limited portions of the study area were categorized as low vulnerability due to their relatively higher elevations. The dominance of low-elevation zones highlights the susceptibility of the mangrove ecosystem to coastal hazards and environmental change.

Mangrove Vulnerability Assessment Map (2025)

All four parameters were integrated using a Weighted Overlay Analysis (WOA) in the QGIS environment. Each input layer was reclassified to a common 1-5 ordinal vulnerability scale prior to overlay. Weights for each layer were assigned based on a combination of expert elicitation and a review of published mangrove vulnerability assessment frameworks (Saaty, 1980; Pourghasemi et al., 2012). The final weights assigned are presented in (Table 1). After the weights assigned, Mangrove vulnerable zones are classified as high vulnerable, vulnerable and low vulnerable (Figure 10).

Result

LULC Classification and Mangrove Transition (2015-2025)

Supervised classification of 2015 and 2025 classifies the satellite data into five LULC classes such as Mangroves, Fallow Land, Crop Land, Sandy Area, and Waterbody (Figure 2, Figure 3). Where the mangrove extraction of 2015 and 2025 is done from the LULC classification (Figure 4). The resulting transition layer was derived from the decadal LULC 2015 - 2025 then it reclassified into Non-Mangrove, Mangrove Stable, Mangrove Loss, Mangrove Gain (Figure 6). Total mangrove extraction of Pichavaram mangrove forest has been increased from 1373.9

ha to 1470.5 ha and determined with the mangrove expansion in terms of area(ha) over 96.6 ha from 2015 to 2025. Post-classification change detection and the resulting transition matrix showed conversions between mangrove and non-mangrove classes (Crop Land, Fallow Land, Waterbody, Sandy Area). Pixels transitioning from Dense Mangrove to non-mangrove classes were classified as loss areas with high vulnerability, while stable or gaining pixels were assigned low vulnerability. The transition layer was reclassified into four categories as Non-Mangrove, Mangrove Stable, Mangrove Loss, and Mangrove Gain.

Spatial Pattern of Mangrove Vegetation Health Using NDVI (2025)

NDVI values derived from 2025 Landsat 8 OLI imagery, clipped to the mangrove extent (Figure 6), were reclassified into five vulnerability classes based on natural breaks. Areas with low NDVI corresponded to canopy thinning, dieback, or degradation and were assigned higher vulnerability, whereas areas with high NDVI reflected dense, vigorous canopy cover and were assigned lower vulnerability.

Patch Area and Core Area Analysis

Patch analysis revealed a reduction in mangrove fragmentation between 2015 and 2025. The total number of patches decreased from 1393 to 932 in 2025 (Figure 7), while mean patch size increased from 0.99 ha to 1.58 ha. The number of Very small patches decreased from 1297 to 852 is due to the genuine ecological coalescence of patches and the area occupied by large and very large patches increased. These results suggest Fragmentation has decreased between 2015 and 2025 and the mangrove landscape appears to have become more consolidated, with fewer tiny patches and larger continuous mangrove blocks. Where Core area analysis revealed an increase in interior mangrove habitat between 2015 and 2025 (Figure 8). The total core area increased from 237.12 ha to 330.32 ha, while the proportion of core habitat increased from 17.26% to 22.46% of the total mangrove area. This increase indicates improved patch integrity and reduced edge influence, suggesting a decrease in fragmentation-related vulnerability within the mangrove ecosystem. These results collectively indicate a shift toward fewer, larger, and more contiguous mangrove patches between 2015 and 2025 which has a reduction in fragmentation, an increase in core habitat area, and improved patch connectivity, all pointing to enhanced ecological resilience of the mangrove ecosystem over the study period.

Fragmentation Vulnerability

Patch class analysis of the 2025 mangrove LULC binary layer showed that smaller patches with irregular shapes and high edge density exhibited greater susceptibility to edge effects. The fragmentation score, reclassified into five vulnerability classes, showed that very small, highly fragmented patches fell into the Very High vulnerability category, indicating high risk of total loss under continued disturbance, while large, contiguous patches fell into the Very Low vulnerability category (Figure 9).

DEM-Based Topographic Vulnerability

Reclassification of the SRTM DEM into five vulnerability classes showed that most of the study area falls within the high and very high vulnerability classes, reflecting predominantly low-lying terrain prone to tidal flooding, storm surges, and sea-level rise. Only a limited portion of the area, associated with relatively higher elevation, fell into the low vulnerability class. This dominance of low-elevation zones underscores the overall susceptibility of the mangrove ecosystem to coastal hazards. The present study is aimed to bring out the Mangrove Vulnerability Assessment of Pichavaram region using various parameter like Mangrove NDVI (2025), Mangrove Transition (LULC 2015-2025), Fragmentation Vulnerability that derived from Patch Area and Core Area Analysis, Digital Elevation Model (DEM) were generated using multi satellite data and other collateral data sets. We have derived the Mangrove vulnerability zones of the study area which shows the classified vulnerable zones.

Mangrove Vulnerability Assessment

With the help of Weighted Overlay Analysis, All the four parameters were reclassified into a 1-5 vulnerability scale before the overlay process. Those parameters are assigned by the weights based on their influence to determine the mangrove vulnerability assessment. The vulnerable zones are identified, and their area coverage shows with 8% of highly vulnerable, 20% of vulnerable, 72 % of low vulnerable zones (Table 2).

Accuracy Assessment and Validation

The LULC classifications for both epochs were validated using a random sampling approach of 100 random sample points with ground-truth reference points derived from the high-

resolution Google Earth imagery interpretation (2015& 2025 data) that resulted in kappa coefficient and the overall accuracy (%) for 2015 as 0.865 and 91%, for 2025 as 0.885 and 92% (Congalton, 1991; Foody, 2002). The final mangrove vulnerability map was qualitatively validated against published fragmentation and degradation studies for Pichavaram, historical records of mangrove loss reported in the Forest Survey of India State of Forest Reports (2015-2021).

Discussion

The Pichavaram Mangrove Forest is one of the most important coastal ecosystems in India and represents the second largest mangrove forest in the country after the Sundarbans. Located along the southeast coast of Tamil Nadu, it serves as a natural barrier that protects coastal communities from cyclones, storm surges, coastal erosion, and sea-level rise (Alongi, 2002; Duke et al., 2007). The dense network of mangrove vegetation and tidal channels helps dissipate wave energy and stabilizes shorelines, thereby reducing the impact of extreme weather events (Kathiresan & Bingham, 2001). Ecologically, Pichavaram supports rich biodiversity by providing habitat, breeding grounds, and nursery areas for numerous species of fish, crabs, prawns, molluscs, birds, and other wildlife (Kathiresan, 2000). The forest also plays a vital role in maintaining coastal food webs and sustaining local fisheries, which are an important source of livelihood for surrounding communities (Kathiresan & Bingham, 2001). In addition, mangroves are highly efficient blue carbon ecosystems, capable of storing significant amounts of carbon in their biomass and sediments, making Pichavaram important for climate change mitigation (Donato et al., 2011).

These ecosystem shows variation and being subject to sustained anthropogenic pressures, has shown a measurable net expansion and structural consolidation between 2015 and 2025 rather than the continued decline reported for many other Indian mangrove systems (Ghosh et al., 2022). That mangrove expansion has been identified through the help of remote sensing techniques and satellite imagery like Landsat 8 OLI imagery from USGS Earth Explorer. The increase in total mangrove area of Pichavaram forest from 1373.9 ha to 1470.5 ha, coupled with a reduction in patch count from 1393 to 932 (Figure 7) and a corresponding rise in mean patch size from 0.99 ha to 1.58 ha, indicates that smaller, isolated fragments have either been lost or have coalesced with adjacent patches over the decade (McGarigal & Marks, 1995; Turner et al., 2001). This pattern is consistent with the broader understanding that fragmentation reduction and the consolidation of patch geometry are strong indicators of

improving ecological resilience, since larger and more contiguous patches are inherently less susceptible to edge-driven degradation (Roy et al., 2025). Thus, the increase of mangrove area in terms of hectare on the Pichavaram mangrove forest over a decade because of the afforestation and mangrove plantation on the defined study area by the organizations and other nearby farmers.

The increase in core habitat area from 237.12 ha (17.26%) in 2015 to 330.32 ha (22.46%) in 2025 reinforces this interpretation (Figure 8), since core area expansion reflects a genuine improvement in interior habitat quality rather than a simple gain in peripheral or marginal vegetation. Interior patches are buffered from edge effects and therefore tend to support denser canopy cover and higher photosynthetic vigour, which is reflected in this study by the correspondence between high NDVI zones and Low vulnerability classes. Conversely, the 8% of the assessed area classified as Highly Vulnerable was concentrated almost exclusively along the forest periphery, adjacent to ponds, agricultural land, and sediment-laden channels (Table 2), where small patch size, high edge density and the integration of increased fragmentation metrics with lower NDVI values results in elevated composite vulnerability scores, indicating greater ecosystem vulnerability (Turner et al., 2001). This spatial pattern, in which vulnerability is concentrated at the interface between the mangrove and surrounding land uses rather than distributed evenly across the forest, and underscores that anthropogenic land-use pressure, rather than purely climatic or hydrological factors, remains the dominant driver of localised degradation at Pichavaram (Alongi, 2002). The dominance of low-lying terrain across the study area, and the resulting classification of most of the landscape into high and very high topographic vulnerability, indicates that the Pichavaram mangrove remains structurally exposed to tidal inundation, storm surge, and longer-term sea-level rise impacts irrespective of its current vegetative health (Nicholls & Cazenave, 2010). This highlights a key limitation of vegetation and structure-based vulnerability indicators in isolation shows a patch may exhibit high NDVI, low fragmentation, and substantial core area, and therefore appear ecologically resilient in the short term, while still being situated in a zone of high physical exposure to climate-driven hazards over longer time horizons. The integration of the DEM layer within the weighted overlay framework was therefore essential for capturing this latent, slower-acting dimension of vulnerability that would not be visible from spectral or landscape-structure data alone (Loveland & Ellison, 2007; Alongi, 2009).

The high classification accuracies obtained for the 2015 and 2025 LULC shows overall accuracy of 91% and 92%, with kappa coefficients of 0.865 and 0.885 respectively.

Nonetheless, several methodological constraints should be acknowledged when interpreting the findings. The use of 30 m resolution Landsat and SRTM data, while suitable for landscape-scale assessment, may not adequately resolve fine-scale patch boundaries, narrow tidal creeks, or small-scale dieback events that finer-resolution sensors such as Sentinel-2 could capture. The weighting scheme used in the Weighted Overlay Analysis, although grounded in established multi-criteria decision analysis principles, inherently carries a degree of subjectivity, and the vulnerability classification could shift under alternative weighting assumptions (Malczewski, 2006). Additionally, the analysis is based on a single recent NDVI epoch and a two date LULC comparison, which captures net decadal change but cannot fully account for interannual variability driven by monsoon intensity, salinity fluctuation, or short-term anthropogenic disturbance events occurring between the two reference years (Alongi, 2015; Wulder et al., 2016). With the help of decadal analysis and mangrove transition shows the expansion of mangrove along with core area expansion leads to the rapid incline of mangrove area of Pichavaram. Future work incorporating higher-resolution multi-temporal imagery, field-validated biomass or canopy-height data, and sensitivity analysis of the overlay weights would help to refine and strengthen the robustness of the vulnerability classification presented here (Alongi, 2009; Simard et al., 2019).

The net gain recorded at Pichavaram over 2015-2025 stands out against a regional backdrop that has generally been characterised by contraction. Continent-scale assessments have repeatedly identified aquaculture conversion, agricultural encroachment, and unregulated coastal development as the principal drivers of mangrove loss across South and Southeast Asia over the past two decades (Richards & Friess, 2016), while earlier satellite-based global inventories similarly flagged large stretches of the Indian coastline as vulnerable to continued attrition (Giri et al., 2011). Viewed against this backdrop, the Pichavaram trajectory of net area gain, reduced fragmentation, and core-area expansion is atypical, and it strengthens the inference that locally targeted afforestation and community-supported plantation drives have measurably outweighed the background pressures that continue to erode comparable systems elsewhere along the east coast (Kathiresan, 2018).

The patch-level consolidation in patch count alongside the rise in mean patch size, is also consistent with findings from very high-resolution fine-scale monitoring of Pichavaram, where plantation driven expansion was documented alongside continuing shoreline erosion and pockets of dieback in older natural stands (Ghosh et al., 2022). That work illustrates a useful caution for the present study, because coarser 30 m Landsat-based mapping tends to generalise

narrow creeks and small dieback patches into surrounding vegetation classes, it can understate the degree of internal heterogeneity within what this study's landscape metrics register as a single, consolidated patch. In other words, the patch and core-area gains reported here most plausibly reflect a genuine net improvement in landscape structure, but they may still be masking small, spatially discrete zones of stress that would only become visible at sub-30 m resolution.

The reliance on a single NDVI epoch to characterise present-day canopy vigour also warrants some caution, since mangrove NDVI in systems such as Pichavaram is known to fluctuate seasonally in response to tidal inundation regime, monsoon timing, and salinity, independent of any longer-term trend in canopy health (Selvam et al., 2019). The methodological approach adopted here, that is, combining Landsat derived LULC classification with a DEM-based topographic layer inside a weighted overlay, follows a well-established line of practice in mangrove remote sensing, where medium-resolution multi-temporal Landsat imagery remains the most widely used data source for tracking decadal-scale mangrove cover change worldwide, owing to its long historical archive and free availability (Vasquez et al., 2023). At the same time, recent reviews of mangrove mapping methods have emphasised that classification accuracy and the reliability of downstream change and vulnerability products depend heavily on the choice of classifier, the quality of training data, and the spectral separability of mangrove from adjacent agricultural land covers (Maurya et al., 2021). The overall accuracies of 91% and 93% obtained for the 2015 and 2025 classifications in this study therefore compare favourably with typical benchmarks reported in the wider literature, and this lends further credibility to the transition statistics on which the vulnerability assessment is built. The pervasiveness of high and very high topographic vulnerability across the study area, largely independent of vegetation condition, suggests that vegetation-based restoration and plantation efforts, however successful in expanding canopy cover, cannot on their own resolve the forest's underlying exposure to sea-level rise and storm surge. A genuinely resilient long-term management strategy for Pichavaram would therefore need to combine continued community-supported afforestation and edge-zone regulation with hydrological and sediment-management interventions capable of addressing the low-lying topographic exposure that vegetation indices alone cannot capture.

Conclusion

This study presents an integrated, GIS-based multi-criteria vulnerability assessment of the Pichavaram mangrove ecosystem, combining mangrove NDVI, decadal LULC transition, fragmentation, core-area, and DEM-derived topographic layers into a single composite vulnerability index through Weighted Overlay Analysis. The results indicate that the Pichavaram mangrove forest expanded by approximately 96.6 ha between 2015 and 2025, growing from 1373.9 ha to 1470.5 ha, while simultaneously becoming structurally less fragmented, with patch count falling from 1393 to 932 and mean patch size rising from 0.99 ha to 1.58 ha. Core habitat area increased correspondingly from 237.12 ha to 330.32 ha, raising the proportion of core habitat from 17.26% to 22.46% of the total mangrove extent. Together, these trends point to a decade of net consolidation and improving structural resilience for this Ramsar-designated mangrove system, in contrast to the degradation trajectories documented for many other mangrove forests in India and elsewhere.

Simultaneously the composite vulnerability map showed that the vulnerability is not uniformly distributed. approximately 8% of the assessed area, concentrated along the forest periphery adjacent to aquaculture ponds, agricultural land, and sediment-prone channels, falls within the Highly Vulnerable class, while 20% is classed as Vulnerable and the remaining 72%, corresponding mainly to dense interior core patches with stable LULC and low fragmentation, is classed as Low Vulnerable. The persistence of high physical exposure indicated by the DEM-based topographic layer further suggests that, despite the encouraging structural and vegetative trends, the mangrove remains inherently susceptible to tidal inundation and sea-level rise over longer time horizons. These findings demonstrate the value of integrating multi-source geospatial data with multi-criteria decision analysis for producing spatially explicit, evidence-based vulnerability maps that distinguish between resilient core habitat and at-risk peripheral zones. The framework developed in this study, integrating NDVI, LULC transition, fragmentation, core-area, and topographic exposure layers within a weighted overlay structure, is readily transferable to other mangrove systems where multi-temporal satellite and elevation data are available, offering a scalable approach to support evidence-based coastal ecosystem management and policy planning under increasing climatic and anthropogenic pressure.

Author contributions TMN - Conceptualization, Methodology, Writing - Original draft preparation, Supervision, Software and Validation; SE - Visualization, Writing and Review.

Data availability Dataset produced from this work shall be made available upon reasonable request from the corresponding author.

Competing interests the authors declare no competing interests.

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Layer / Parameter	Data Source	Weight (%)	Rationale
Mangrove NDVI (2025)	Landsat 8 OLI	30	Current health/vigour- direct biophysical indicator
Mangrove Transition (LULC 2015–2025)	Landsat 8 OLI	30	Historical loss/gain- shows active change pressure

Fragmentation Vulnerability	Patch analysis & Fragmentation Score	20	Structural integrity, edge effects, patch isolation
Digital Elevation Model (DEM)	SRTM 30 m	20	Identifies terrain elevation

Table 1. Parameter, data sources, and weights used in the Weighted Overlay Analysis.

Vulnerability Class	Area Covered (%)	Characteristic Features
Highly Vulnerable	8%	Highly degraded patches, high fragmentation, Low NDVI
Vulnerable	20%	Transitional areas, moderate patch isolation
Low Vulnerable	72%	Dense core areas, stable LULC, low fragmentation

Table 2. Mangrove Vulnerability Zones and spatial distribution.

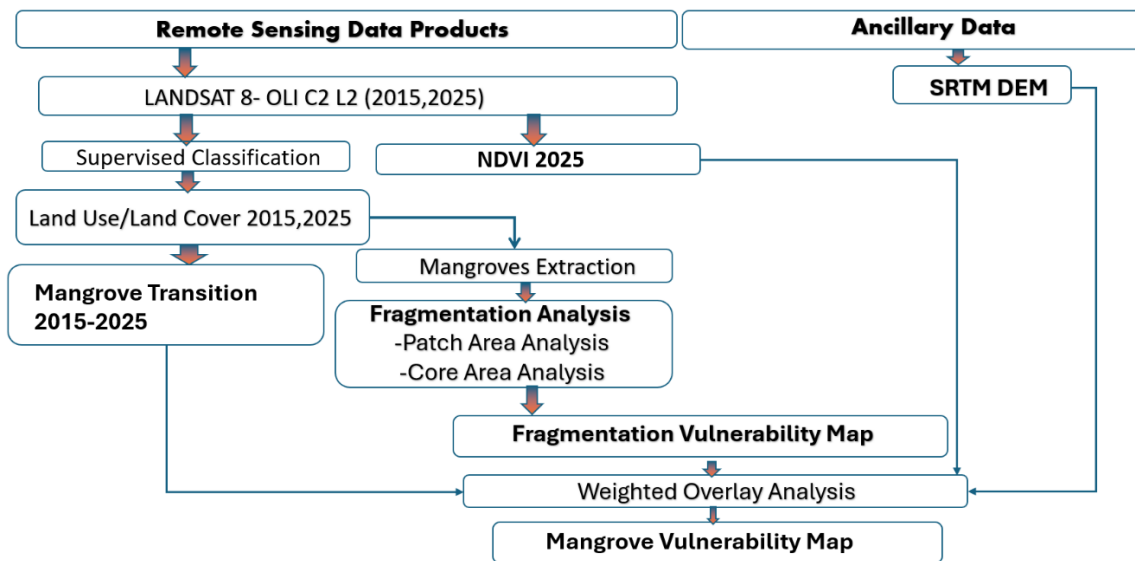


Figure 1: Methodology

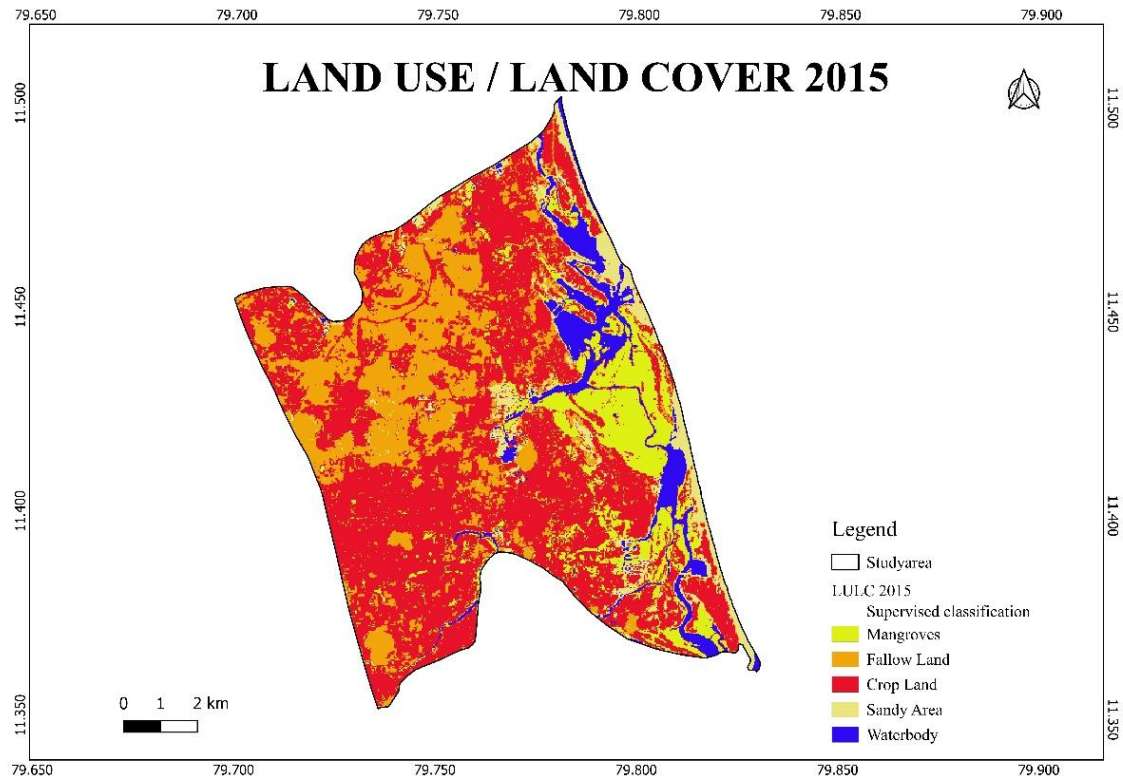


Figure 2: Map showing the Land Use and Land Cover of 2015

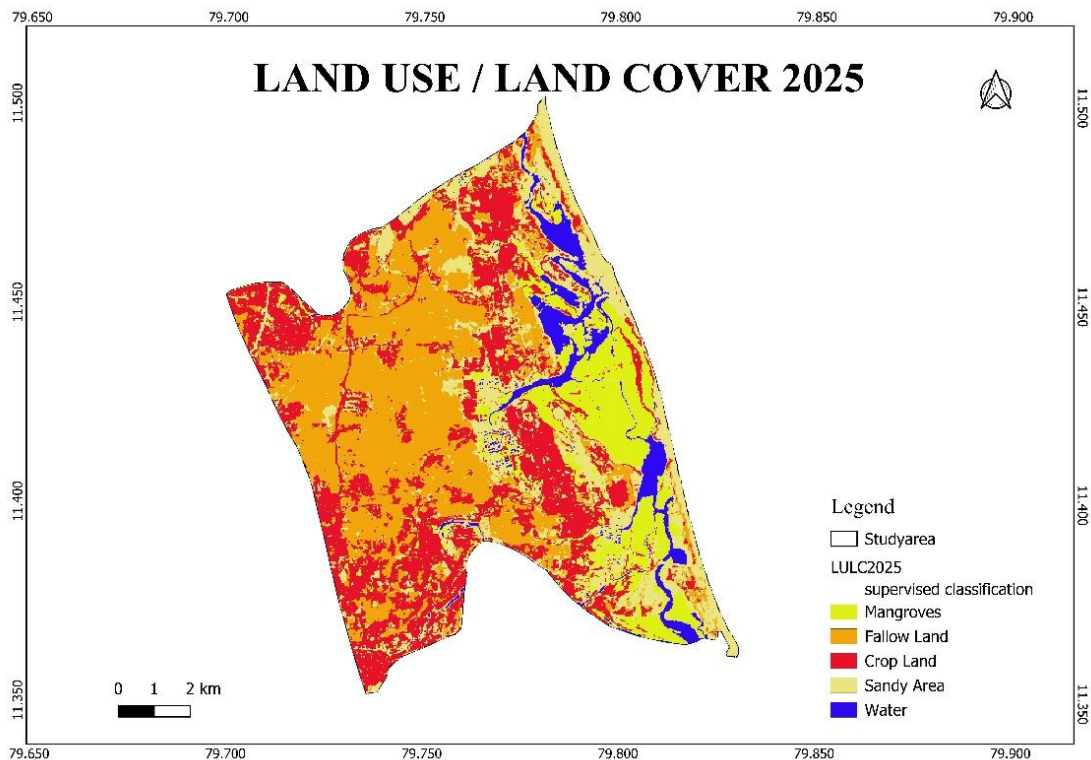


Figure 3: Map showing the Land Use and Land Cover of 2025

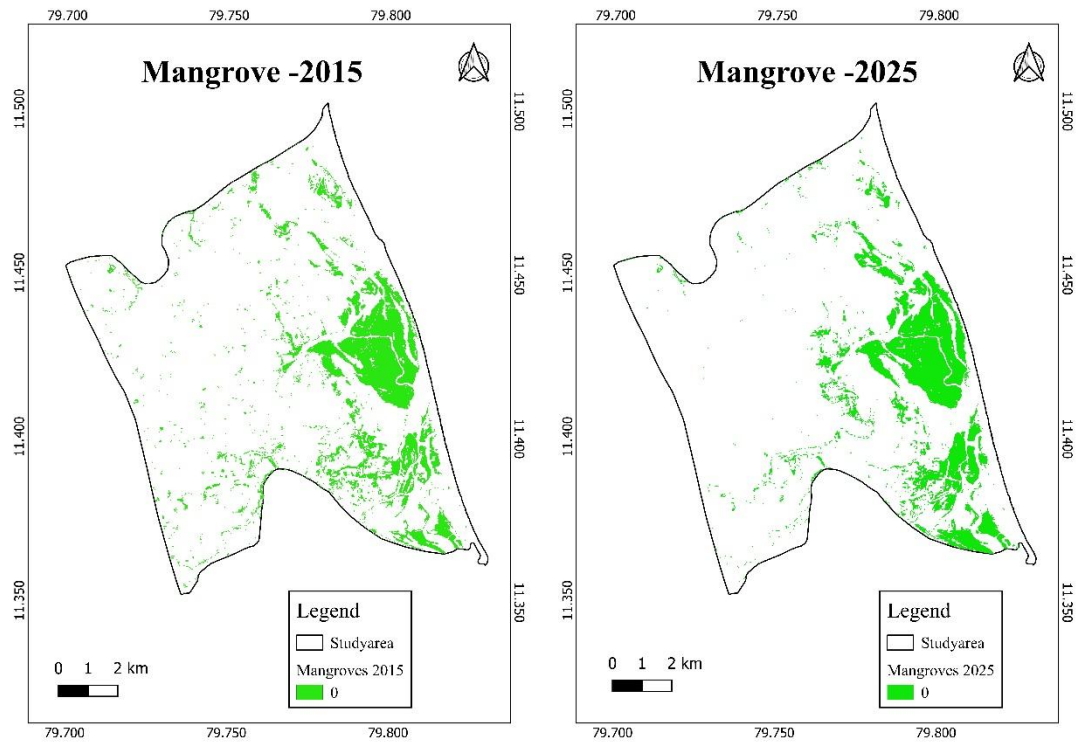


Figure 4: Map showing the extracted Mangrove of 2015 and 2025 from LULC

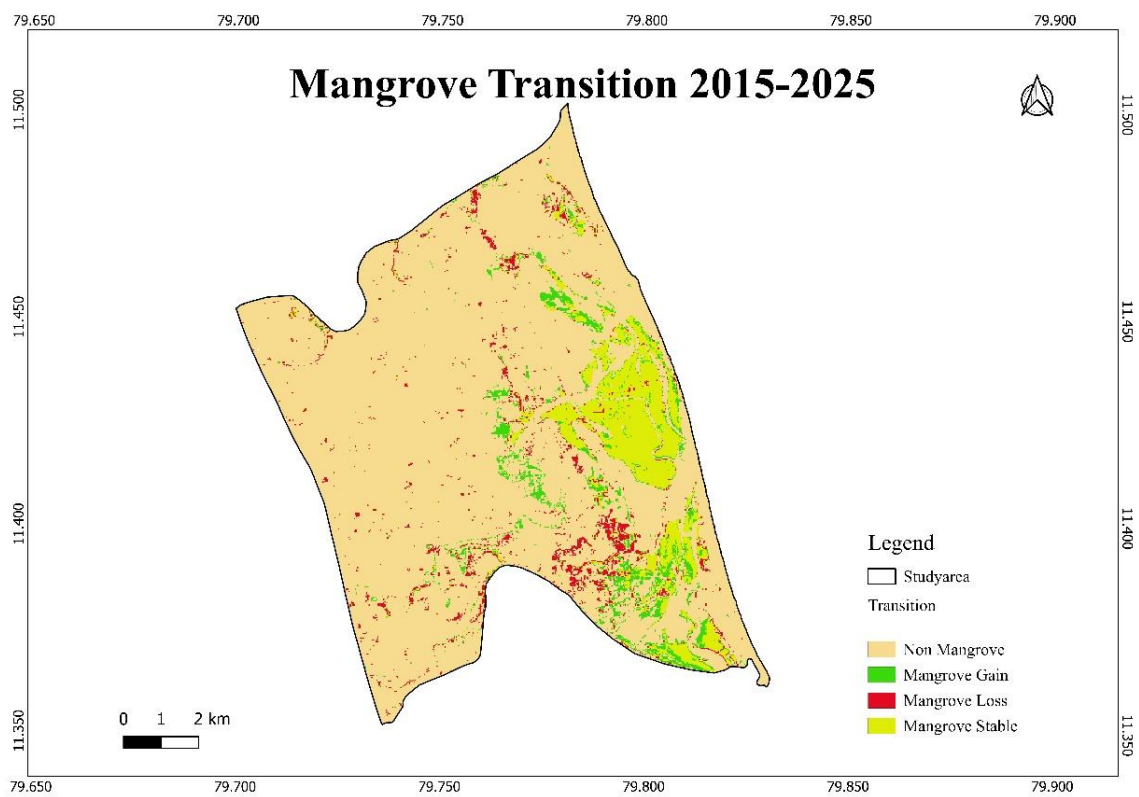


Figure 5: Map showing the Mangrove Transition from 2015 - 2025

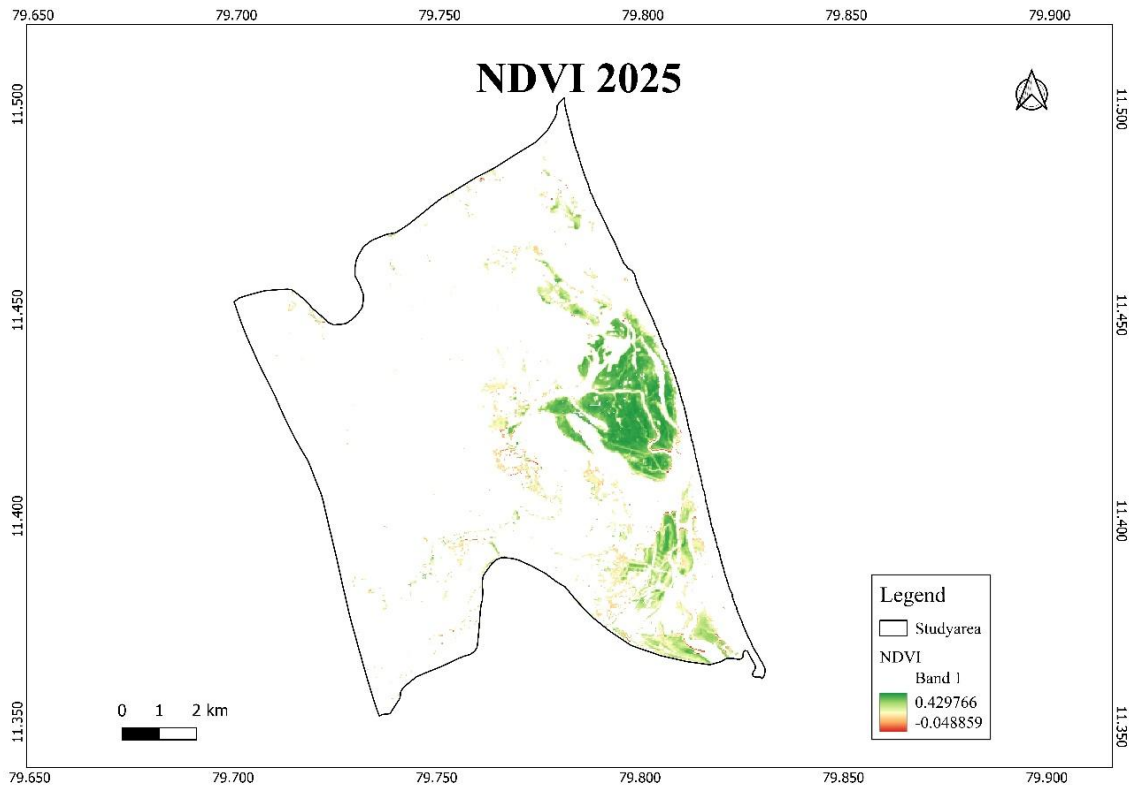


Figure 6: Map showing the NDVI of Pichavaram Mangrove 2025

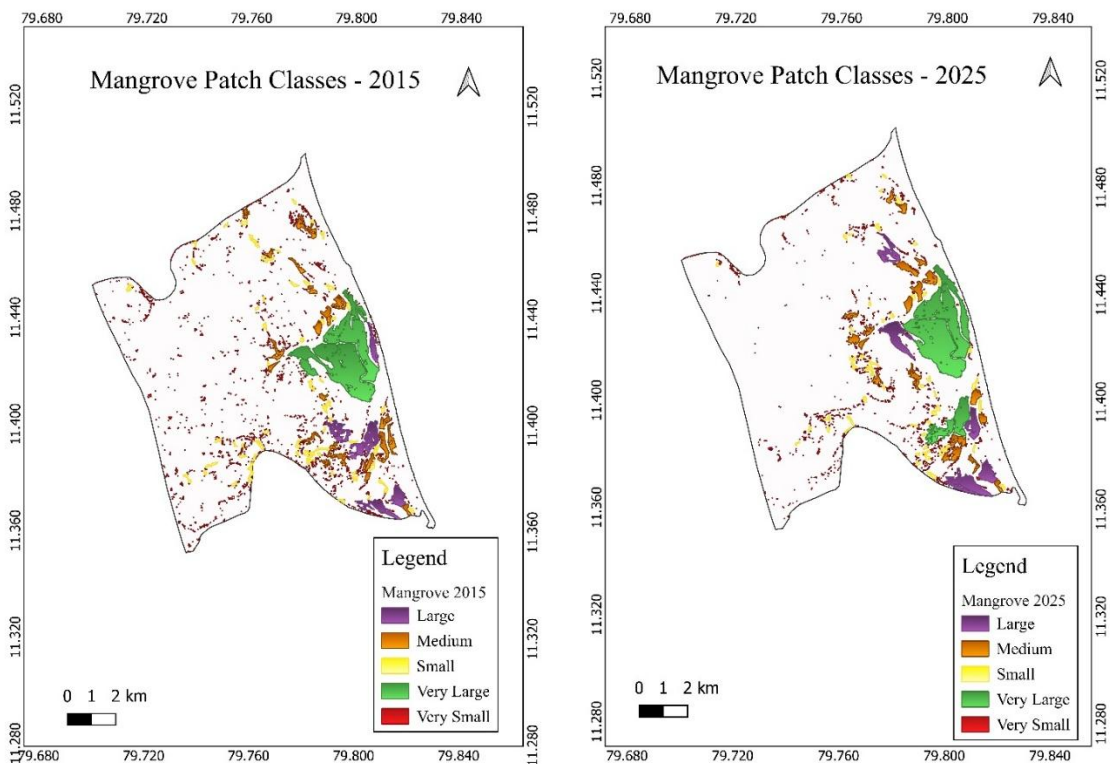


Figure 7: Map Showing the Mangrove Patch Classes of 2015 & 2025

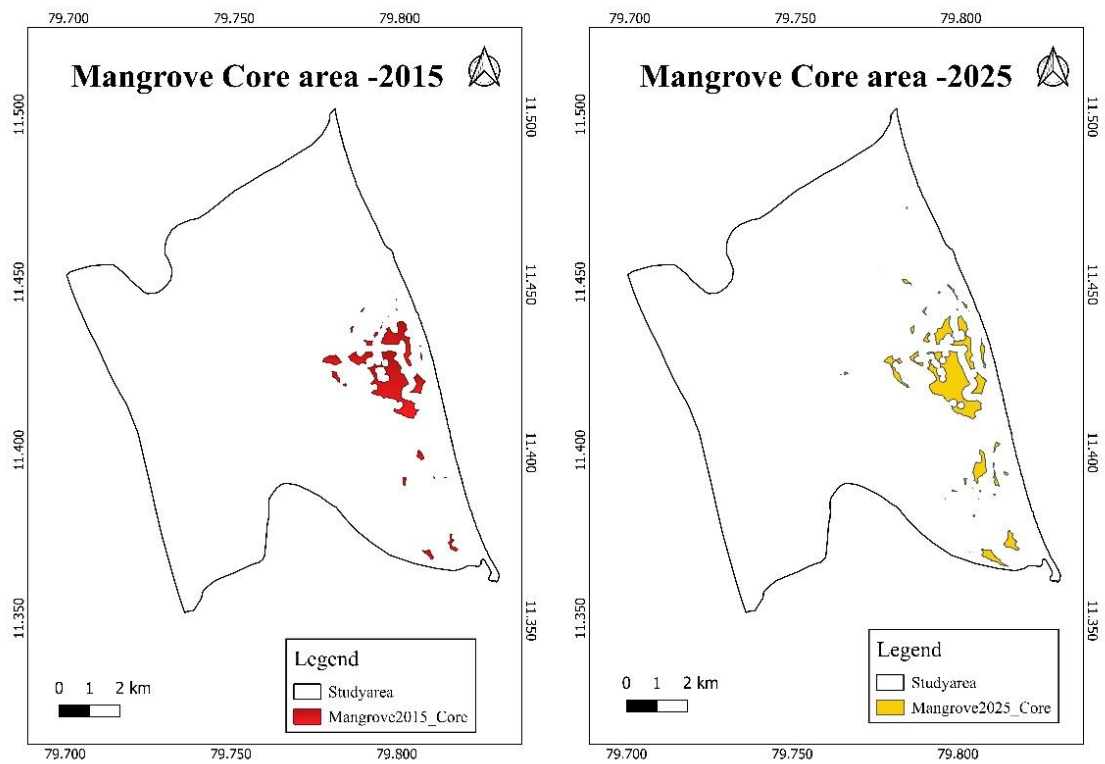


Figure 8: Map Showing the Mangrove Core Area of 2015 & 2025

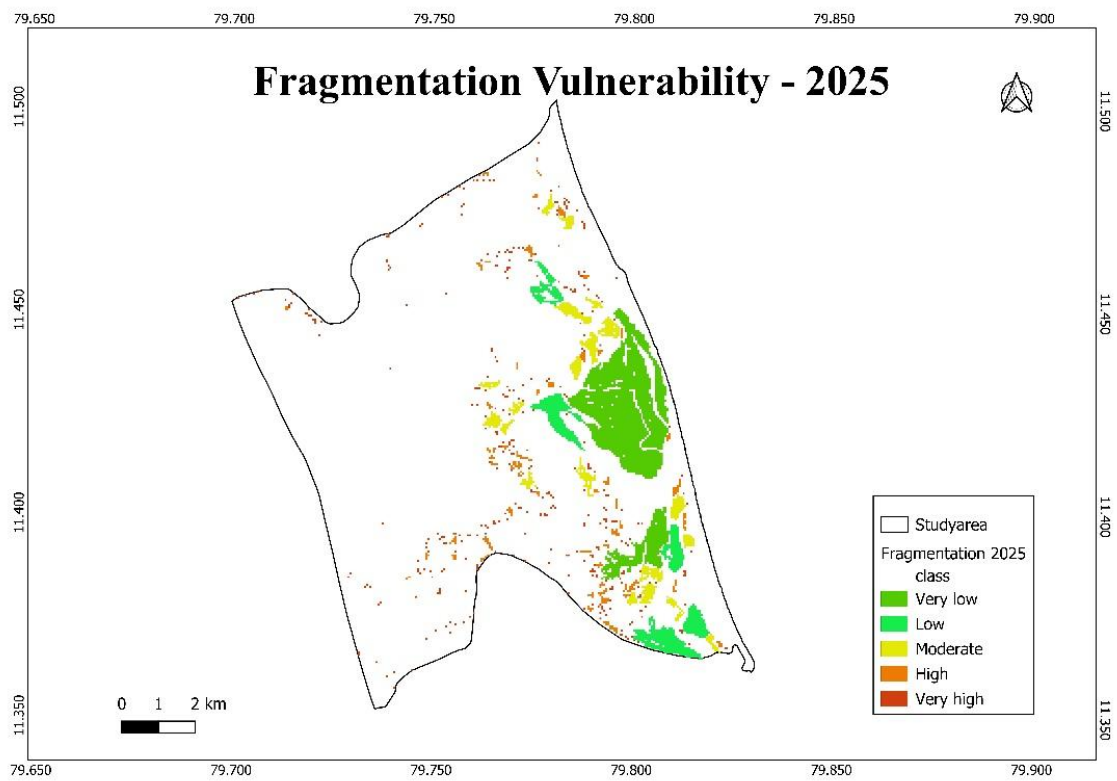


Figure 9: Map showing the Fragmentation Vulnerability of 2025

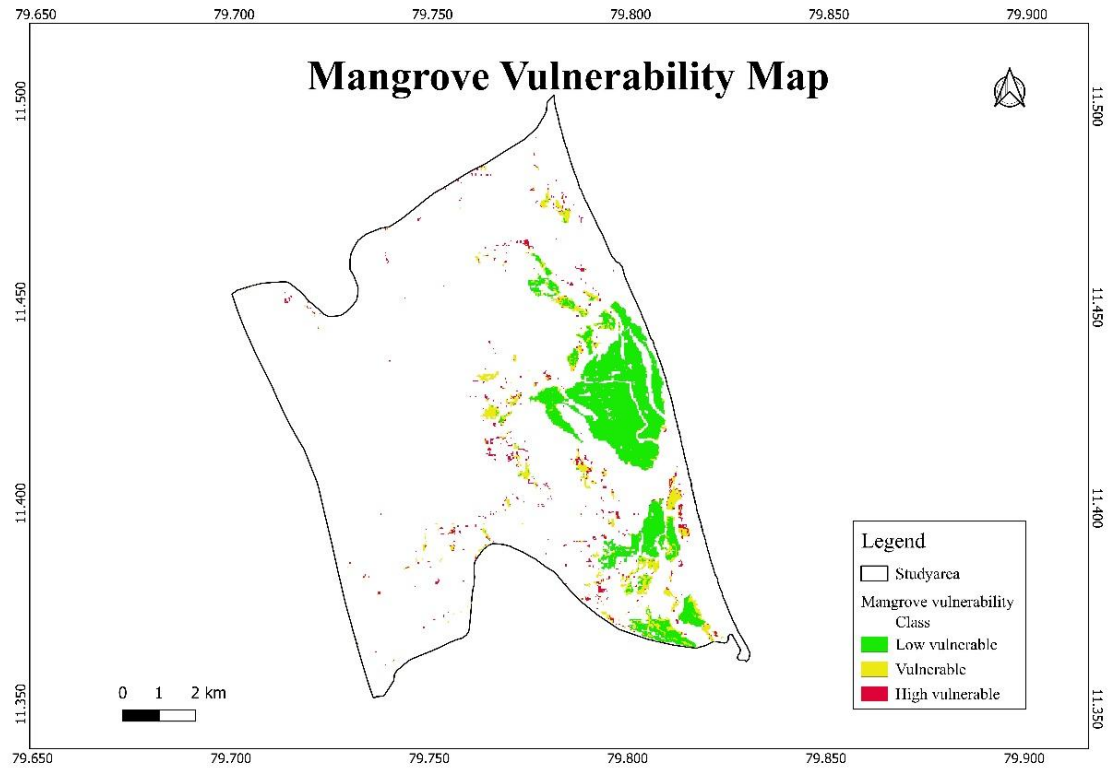


Figure 10: Map showing the Mangrove Vulnerability of 2025